Usman Ali

Ai Engineer – Machine Learning, Backend and Web

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Summary

High impact **AI Engineer** holding a **BS in Artificial Intelligence** with demonstrated production success designing and shipping **Large Language Model (LLM) applications, Retrieval-Augmented Generation (RAG) pipelines, autonomous AI agents, and resilient full-stack web platforms**. Proven record co-founding **ReplyPops** (live commercial product), engineering predictive analytics microservices serving **100+ daily financial traders** at XEN Trades Group, and building custom model evaluation frameworks that identified hidden sub-segment accuracy regressions prior to release. Certified in **AWS Agentic AI** and **Claude Architecture**. Adept at bridging modern AI research with scalable, high-throughput web and cloud backends.

Skills

Language models: Claude, GPT-4o, Llama 3, Gemini. Prompt writing, structured JSON output with Pydantic, context caching, token cost control.

Search and retrieval: Document search combining keyword (BM25) and vector search. ChromaDB, Pinecone, pgvector. Result re-ranking, source citations, reducing wrong answers.

Automation and chatbots: Multi-step workflows, function calling, conversation memory, n8n, human review steps before actions go out.

Machine learning: PyTorch, scikit-learn, TensorFlow. Model testing (precision, recall, ROC-AUC), per-segment accuracy checks, time series.

Web development: React, Next.js, TypeScript, JavaScript, Python (FastAPI, Flask), Node.js, C# / .NET Core 8, REST and gRPC APIs, WebSockets.

Databases and infrastructure: PostgreSQL, ClickHouse, MongoDB, DynamoDB, Redis, RabbitMQ, Redpanda / Kafka, AWS, Docker, Git, CI/CD.

Experience

Co-Founder and Engineer – ReplyPops (replypops.com)

2026 – Present · Islamabad, Pakistan

- Designed and built the ReplyPops product from first sketch to public launch, including business registration and customer onboarding.
- Built the pipelines that write and classify reply drafts using the Claude and OpenAI APIs, with a fixed JSON format for every response. Draft generation became 65% faster, with consistent tone.
- Built the web app with Next.js, TypeScript and REST backends. Shipped weekly updates based on customer feedback.

Trading Analyst and Software / ML Engineer – XEN Trades Group

2024 – 2026 · Remote

- Built and maintained a live analytics and risk dashboard used daily by more than 100 professional traders.
- Wrote services in C#, .NET Core 8, Python and gRPC, using RabbitMQ to stream live order book data.
- Designed database tables in PostgreSQL, ClickHouse and MongoDB and rewrote heavy queries so dashboards load in under a second.

- Automated market data collection and wrote integration and regression tests.

Data Analysis and Research Intern – EcoServants (Ecological Servants Project)

Remote

- Queried, cleaned and checked large ecological datasets in SQL and built time-series tables for the Community Site Report.
- Built a web dashboard in JavaScript and PHP that lets field staff without technical backgrounds rank intervention sites by expected impact.

Technical Director – Mega Code War 4.0, HITEC University

Taxila, Pakistan

- Built the registration system and data pipelines for Pakistan's largest university coding competition (360+ registered participants) and ran the servers on competition day with no downtime.

Projects

Lead Scoring and Routing Workflow (n8n, Claude API, webhooks, Python) Checks incoming webhook data, scores each lead from 1 to 10, writes a personalised follow-up email, and sends high-scoring leads to sales after a person approves.

TickerLake – Live Market Data Pipeline (Python, Redpanda / Kafka, ClickHouse, WebSockets) Collects live order book and trade data from Binance and Coinbase into ClickHouse. Removes duplicates, sends failed messages to a separate queue, and runs automated tests continuously.

Company Knowledge Base Chatbot (Python, FastAPI, vector embeddings, React) Answers questions from company documents using keyword and vector search, re-ranks results, and shows source links. Cut wrong answers by more than 85%.

Education

BS in Artificial Intelligence – HITEC University, Taxila, Pakistan

Certifications and Awards

- Agentic AI on AWS – certificate
- Claude Architecture and Prompt Engineering – certification
- 11th place – National competition, NUST
- 1st place – Inter-department coding competition, HITEC University
- Certified Web Development Trainer – Anam Foundation
- Huawei cloud - Certification in progress